

Vaughn Campos

Brooklyn, NY | (845) 269-7031 | vcampos4545@gmail.com | [GitHub](#) | [LinkedIn](#)

SUMMARY

Full-stack software engineer with professional experience building production web and mobile applications, plus hands-on experience designing embedded firmware and robotics control systems. Comfortable working across the entire stack, from React/Next.js interfaces and cloud infrastructure down to low-level C firmware and real-time control loops, with additional experience applying LLMs in production.

WORK EXPERIENCE

SmallTalk (Jelly Jelly), New York, NY, Full Stack Software Engineer

December 2023 – October 2025

- Built the video capture and distribution pipeline for a live social video app: recordings captured locally on iOS or via LiveKit's WebRTC infrastructure, uploaded to S3, encoded into multiple quality renditions through a queued FFmpeg job, and distributed as HLS through a CDN
- Led end-to-end development of a Swift-based iOS application, successfully launching on the App Store to expand the platform's mobile user base
- Architected scalable infrastructure using AWS services (S3, Lambda, EC2), deployed via Vercel, and integrated Supabase for real-time backend services

Orenda Inc., New York, NY, Machine Learning Engineer Intern

June 2022 – August 2022

- Deployed containerized data collection services on Kubernetes to web scrape data into SQL databases
- Built dashboards for the operations team to evaluate site selection for battery assets
- Tuned and deployed LSTM deep learning models to forecast energy demand for the NYC area, informing trading decisions in wholesale energy markets, and improving accuracy by ~10%

PROJECTS

Constellation Design Simulation — C++, Python

2024

- Architected a modular C++/Python simulation framework (propagation core, analysis layer, optimization driver) with unit-tested components, enabling reliable, repeatable parameter sweeps across thousands of orbital design candidates
- Implemented a high-fidelity orbit propagator modeling J2 oblateness, atmospheric drag (USSA76 density model), solar radiation pressure, and third-body gravity from Sun/Moon ephemerides, combined with Walker-pattern constellation generation to keep long-duration coverage sweeps physically accurate
- Built a Monte Carlo trade study tool with parallelized sweeps across CPU cores, surfacing Pareto-efficient constellation designs across multi-dimensional coverage and cost trade-offs

Spacecraft Dynamics Simulation — C++

2024

- Built a general-purpose rigid-body physics engine from scratch in C++, including GJK/EPA collision detection and a Sequential-Impulse constraint solver supporting 5 joint types
- Implemented and compared multiple attitude-control strategies (PID, LQR, and cascaded rate/attitude) for a simulated CubeSat using reaction-wheel torque allocation, plus gimbaled thrust-vector control for a multi-body rocket scenario with in-flight staging
- Integrated the engine with a custom OpenGL rendering library I built separately, using an orbital camera and logarithmic depth buffering to visualize scenes spanning both CubeSat and multi-kilometer rocket scales in real time

Calendar Copilot — Next.js, OpenAI, Google Calendar API

2026

- Built an AI calendar assistant using OpenAI function calling in a tool-calling loop, integrating Google Calendar, Gmail, and third-party flight/weather APIs
- Integrated third-party trip-planning APIs (Amadeus flight search, Open-Meteo weather) with city-to-airport resolution and token caching to support end-to-end itinerary planning in the same conversation
- Designed an end-to-end trip-planning flow, flight search plus weather-informed packing lists, that collapses trip-prep anxiety (juggling flight sites, weather apps, and packing decisions separately) into a single conversation

Reaction Wheel Balancer — ESP32, C++, SimpleFOC

2025

- Designed and built a reaction-wheel-balancing robot, implementing IMU sensor fusion, field-oriented motor control, and a custom state-feedback control law on an ESP32
- Wrote embedded C firmware with a hardware watchdog and fault recovery for I2C bus lockups, improving reliability during extended autonomous runs
- Validated control gains via LQR design in Python and tuned live over serial, achieving stable closed-loop balancing under manual disturbances

EDUCATION

Cornell University, College of Engineering, Ithaca, NY

August 2019 – May 2023

B.S. Computer Science, Operations Research and Engineering

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, C/C++, Swift

Web & Cloud: React, Next.js, Node.js, AWS, GCP, Vercel, Docker, Kubernetes

Databases: PostgreSQL, MongoDB, Pinecone (vector search)

AI/ML: OpenAI API, LLM agents & tool calling, RAG, PyTorch, Claude Code (agentic development workflows)

Embedded & Controls: ESP32, STM32, bare-metal/register-level C, FreeRTOS, I2C/SPI/UART, control theory, sensor fusion

Hardware & Fabrication: Fusion 360, TinkerCAD, 3D printing, soldering, bench instrumentation (multimeter, oscilloscope, logic analyzer)